SIMPLEX ARGUS Project: MCR Report

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1 General Project Information

Last updated	7 October 2026
Programme	M2 Newspace
Team / project name	DeOrbit (ARGUS project)
Project coordinators	Adnane Ait Zidane and Thomas Weber
Coordinators' email addresses	adnaneaitzidane@outlook.com – thweber@outlook.fr

2 Project Overview

The ARGUS project aims to detect a piece of space debris and then modify its trajectory using a laser, with the entire system hosted on a 6U nanosatellite. The concept rests on three interdependent elements : the **design** of the debris and the satelite, **debris detection**,**laser-induced trajectory modification**,and its complete instrument suite.

This is an experimental project : a satellite of this size cannot generate high power levels. It nevertheless provides a means of testing the feasibility of modifying the trajectory of debris using a space-based laser. The velocity changes achievable with this technology would be of the order of metres per second. Figure 1 shows the mission timeline.

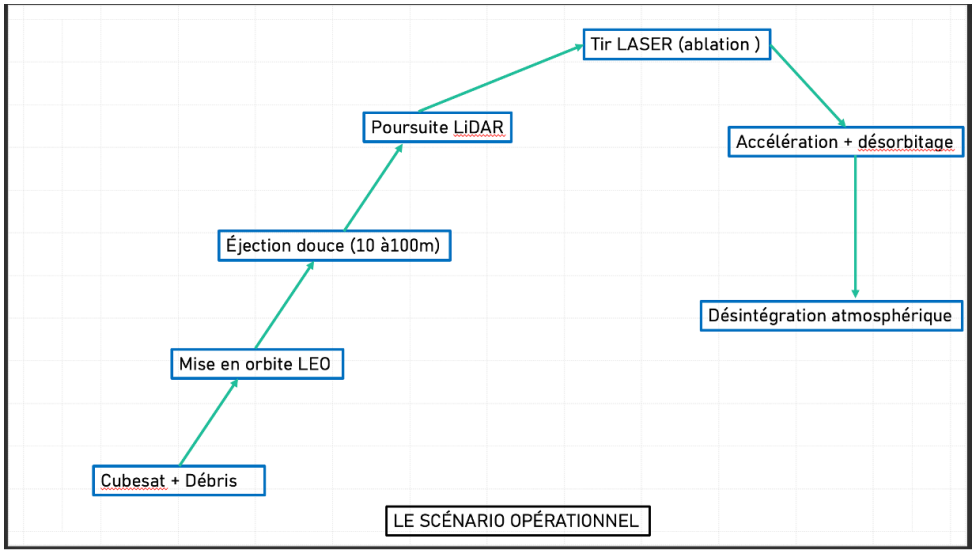


FIGURE 1 – ARGUS mission timeline

9.1 Mission 5×5 Risk Matrix

Risk ID	Risk Title & Description	L	C	Mitigation Strategy	Owner
R-01	**EPM Release Stiction :** Residual flux or mechanical stiction prevents target release.	2	3	Dual-wound coil with 200% reverse demagnetization current ; Delrin guide pins ; TVAC bakeout testing.	Mechanisms Lead

Risk ID	Risk Title & Description	L	C	Mitigation Strategy	Owner
R-02	**Premature Target Release :** Launch vibration separates target from dispenser.	2	4	Magnetic retention force rated > 18 N (> 100 g holding) ; mechanical retaining lips block lateral shear.	Structures Lead
R-03	**Solar Storm Rapid Reentry :** Elevated solar flux ($F_{10.7} > 220$) reduces orbit life to < 30 days.	4	2	350 km baseline provides 3.5-month margin ; expedited CONOPS executes laser firing within 18 days of launch.	Mission Design Lead
R-04	**Target Tracking Dropout :** Tumbling coupon causes optical tracking fades at standoff.	3	3	Primary tracking utilizes onboard lidar and micro-CCR ; 2 cm sphere carried as active challenger.	GNC Lead
R-05	**Optics Plume Contamination :** Ablated plume deposits material on beam director primary mirror.	3	2	Standoff maintained at 300–500 m ; plume expansion density falls off as $1/R^2$, dropping to zero deposition beyond 10 m.	Payload Lead

3 Project Feasibility

This section assesses the feasibility of the ARGUS mission. To do so, we need to calculate certain key parameters essential to successful mission execution. These calculations should also validate our various parameter choices, particularly those relating to the selected LASER sources, the debris material and the target orbit.

3.1 Theoretical Calculation of Selected LASER Parameters

— **Ablation Threshold and Optimal Fluence (F)**
The fluence of our LASER at a wavelength of $\lambda = 1064nm$ with pulses of **nanosecond duration (ns)** represents the energy deposited by this laser per unit area. It is expressed in J/cm^2. The scientific literature places the theoretical ablation threshold between $F_{th} \approx 1J/cm^2$ and $F_{th} \approx 2J/cm^2$ (Philippe class reference). However, to obtain the required optimal propulsive effect in our case, we need to reach the maximum coupling fluence $F_{couplage}$ defined by :

$$\boxed{F_{couplage} \approx 2.5F_{th} \approx 5J/cm^2} \tag{1}$$

for $F_{th} = 2J/cm^2$

— **Pulse energy E_P and peak power P_c**
Pulse energy represents the total energy contained in a single LASER pulse. It is calculated using the following formula :

$$\boxed{E_P = S * F_{couplage}} \tag{2}$$

S represents the area illuminated by the LASER pulse. For a selected focal spot diameter

Likelihood \ Consequence	1 (Negligible)	2 (Minor)	3 (Moderate)	4 (Critical)	5 (Catastrophic)
5 (Very High)	—	—	—	—	—
4 (High)	—	**R-03**	—	—	—
3 (Moderate)	—	**R-05**	**R-04**	—	—
2 (Low)	—	—	**R-01**	**R-02**	—
1 (Very Low)	—	—	—	—	—

of $d = 0.3cm$ and assuming a spherical spot, we obtain the value of S :

$$\boxed{S = \frac{\pi d^2}{4} = \frac{\pi * (0.3)^2}{4} \approx 0.071cm^2} \tag{3}$$

With this value of S, we finally obtain :

$$\boxed{E_P = 0.071 * 5 = 0.35J} \tag{4}$$

Thus, with this value of E_P, we can determine the instantaneous peak power P_c generated by our LASER. This power represents the energy flow per unit time. It is the maximum power reached by the LASER within the pulse itself. It is obtained by dividing the pulse energy by the pulse duration.

$$\boxed{P_c = \frac{E_P}{\tau} = \frac{0.35}{5.10^{-9}} = 70MW} \tag{5}$$

— **Repetition rate f_{rep} and average power P_{moy}**
To accommodate the payload within a $6U$ CubeSat, we assume an average power of $P_{moy} = 1000W$. This value is not arbitrary ; it is primarily dictated by the thermal dissipation limit of a $6U$ CubeSat [1, 2].
With this value, the pulse repetition rate is calculated using the formula below :

$$\boxed{f_{rep} = \frac{P_{moy}}{E_P} = \frac{1000W}{0.35J} = 2857Hz \approx 3KHz} \tag{6}$$

3.2 Calculation of ΔV and Deorbiting Dynamics

The calculation of the required velocity change ΔV and the transferred impulse is based on three main elements :

— **The momentum coupling coefficient (C_m)**
When a LASER beam interacts with the surface of a material at a fluence above its ablation threshold (aluminium in this case), the material vaporises instantaneously. This ejection of matter, the **rocket effect**, generates a recoil force. This coefficient measures the efficiency with which the LASER transfers momentum to the debris. The coupling is defined by the following fundamental relationship [2] :

$$\boxed{C_m = \frac{I}{E_{LASER}} = \frac{\Delta p}{E_{LASER}}} \tag{7}$$

with **I** the total generated impulse ($N.S$) and E_{LASER} the total absorbed LASER energy in (**J**). C_m is expressed in ($N.s/J$).
According to the experimental data in [2] for a nanosecond LASER at a wavelength of $\lambda = 1064nm$, the stabilised value at peak fluence is equal to $C_m = 30\mu N.s/J$

— **Average generated force** F_{moy}
This force is therefore given by the expression below :

$$\boxed{F_{moy} = c_m * P_{moy} = 30.10^{-6} * 1000 = 0.03N = 3mN} \tag{8}$$

— **Orbital dynamics**
In our case, we opt for a staged atmospheric re-entry, lowering the debris perigee and allowing the atmosphere to complete the process. We therefore apply our LASER impulse directly opposite to the velocity vector $\vec{v_{debris}}$ of the debris at apogee.
According to the vis-viva equation and Kepler's laws, we define the circular velocity $\vec{v_c}$ of the debris in its initial orbit (at $350Km$). This is the velocity required for the centrifugal force to balance Earth's gravity exactly and maintain its trajectory.

$$\boxed{v_c = \sqrt{\frac{\mu}{R_E + h}} = \sqrt{\frac{3.986.10^{14}}{6371000 + 350000}} = 7701m/s} \tag{9}$$

To lower its perigee to a critical altitude of $h_p = 100Km$, the velocity at the new apogee $\vec{v_a}$ is given by :

$$\boxed{v_a = \sqrt{\mu\left(\frac{2}{r_a} - \frac{2}{r_a + r_P}\right)} = \sqrt{3.986.10^{14}\left(\frac{2}{6371000 + 350000} - \frac{2}{6721000 + 6471000}\right)} = 7628m/s} \tag{10}$$

Here, $r_a = R_E + h$, represents the apogee radius of the new trajectory ;
$r_P = R_E + h_P$, represents the perigee radius of the new trajectory.
We select a LEO orbit at $h = 350Km$, which is the initial orbit of our debris (corresponding to the initial orbit of our CubeSat).
$h_P = 100Km$ corresponds to the target altitude of the debris. At this orbit, the atmosphere is sufficiently dense to decelerate our debris and cause re-entry.
$\mu = 3.986.10^{14}m^3/s^2$ is Earth's standard gravitational parameter.
$R_E \approx 6371Km$ represents Earth's mean radius.
The expression for ΔV is therefore written as follows :

$$\boxed{\Delta V = v_c - v_a = 7701 - 7628 = 73m/s} \tag{11}$$

Numerical result : $\boxed{\Delta V = 73m/s}$.
The total required firing duration can also be calculated.

$$\boxed{\Delta t = \frac{m_{debris} * \Delta V}{F_{moy}} = \frac{1 * 73}{0.03} = 2433s} \tag{12}$$

The required firing duration Δt is approximately 40 minutes and 33 seconds to impart a velocity change of $\Delta V = 73\ m/s$ to our debris, divert it from its initial orbit and cause atmospheric re-entry. [3]

But it's impossible to keep the contact between the debris and the CubeSat for 40 minutes. It's no nessecary to create a big ΔV if you change the orbit with a small ΔV and wait for the debris to re-enter the atmosphere it's a success. Thanks to this experimental project, we will be able to validate the feasibility of our concept and the effectiveness of our LASER ablation system in space.

4 Design and Debris Management

The shapes of both the nanosatellite and the debris must be selected. The shape of the debris has a major influence on atmospheric re-entry. The means of providing the required power (batteries and solar panels) and managing overheating must be selected according to mission requirements.

— **Nanosatellite geometry** : the structure is designed around a 6U format. The dimensions under consideration are $30 \times 20 \times 10$ cm or $60 \times 10 \times 10$ cm, although other configurations remain possible.

The two next parts of the report describe the LiDAR and laser systems, and the machine learning algorithms used for debris detection and tracking. The both part is the main instrumentation of the nanosatellite, and the most critical for the success of the mission.

5 LiDAR and Laser

Sizing a LASER-ablation deorbiting system involves a direct physical trade-off : the LASER must not destroy the debris through excessive heat input, but must vaporise a very thin layer of its surface to create a propulsive plasma that slows the moving debris. Thus, modifying the orbit of an object (debris) relies on converting the optical energy deposited by laser pulses into the mechanical energy of the plasma jet.

5.1 LASER Ablation Propulsion

When a high-intensity LASER beam strikes the surface of a material, the optical energy is absorbed instantaneously. The temperature of the surface layer rises so rapidly that it undergoes sudden sublimation and becomes a highly energetic plasma. This jet of matter is expelled at very high speeds. Under the physical law of conservation of matter, the ejection of matter generates a mechanical recoil force on the target object [4].

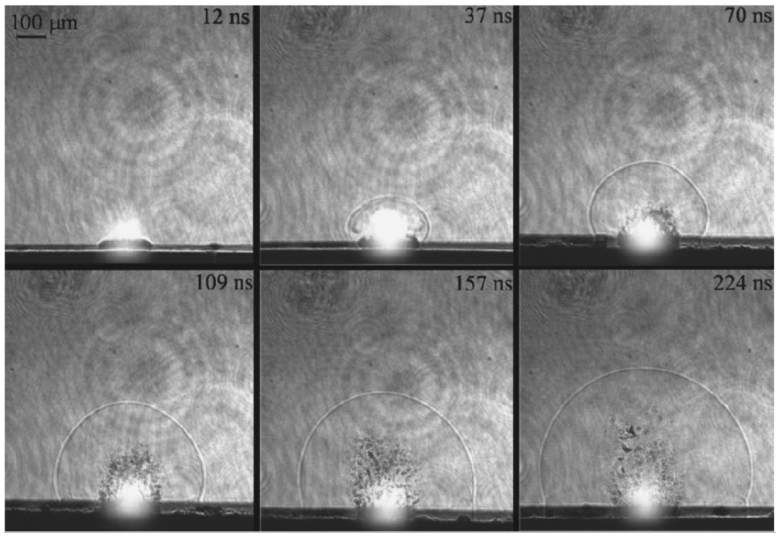


FIGURE 2 – Images of the ablation propulsion process at different times during the laser–aluminium interaction for $\lambda = 1064nm$ and $\tau = 5ns$. The propagation of the shock wave over time can be observed. [4]

5.2 High-Power LASER Characteristics

In this section, we describe the characteristics of our onboard high-power LASER. Selecting the high-power LASER is crucial : if its power or energy is unsuitable, it will probably not be possible to initiate the micropropulsion process at the surface of our target (debris).

1. **Choice of LASER type :**
   We choose a pulsed LASER rather than a continuous-wave laser. The factors determining this choice are summarised in the table below

TABLE 2 – Comparison between continuous-wave and pulsed lasers.

Characteristic	**Continuous-wave laser**	**Pulsed laser**
Effect on debris	Thermal melting	Plasma ablation (mechanical rocket effect)
Peak power	Equal to average power	Several megawatts during the pulse
Generated thrust	Almost zero at this power level	Optimal and directional
Risks	High risk of breaking up the debris	Safe, targeted surface action

2. **Required LASER parameters :**
   The wavelength of our high-power LASER is set to λ =1064 nm in the infrared, with a pulse duration of **nanoseconds (ns)**. This choice is primarily based on the absorption characteristics of the material making up our debris, namely aluminium (Al).

   For this CubeSat project using a $6U$ platform firing a LASER at short range, the optimal average power required for our small aluminium debris object, measuring 1 cm à 10 cm and weighing $1kg$ is approximately $1000W$. The choice of this operating power is deliberate

and results from a specific trade-off validated by orbital mechanics and the physics of laser–matter interaction. The section on project feasibility discusses in detail the importance of reaching this optimal LASER ablation power.

TABLE 3 – Ablation laser parameters.

Parameter	**Value**
Wavelength (λ)	1064 nm (infrared)
Operating mode	Pulsed
Pulse duration (τ)	Nanosecond (ns)
Average power (P_{moy})	1000 W (1 kW)
Repetition rate (f)	3000 Hz (2 kHz)
Pulse energy (E_p)	0,5 J
Peak power ($P_{\text{crête}}$)	100 MW
Ablation threshold fluence	1 J cm^{-2} à 2 J cm^{-2}
Optimal coupling fluence	5 J cm^{-2}

5.3 LiDAR for Detection

To meet the imposed dimensional constraints, we favour a micro-LiDAR over a conventional LiDAR, which would entail higher mass and energy demands. The micro-LiDAR provides optical metrology of the debris : detection, relative position and relative velocity [5]. It consists of three main elements :

— The emission source, which in our case could be a VCSEL (Vertical Cavity Surface Emitting LASER). These are arrays of miniature semiconductor LASERs. The table below summarises the parameters we consider optimal for defining the LASER source of our micro-LiDAR.

TABLE 4 – Micro-LiDAR laser parameters [6].

Parameter	**Specified value**	**Purpose in space**
Wavelength	1550 nm	Prevents blinding by the ablation laser (1064 nm)
Source type	Pulsed laser diode (InGaAs)	Robustness, extreme compactness
Pulse duration	3 ns	Millimetre-level ranging accuracy
Frequency	20 kHz	Smooth real-time target tracking
Pulse energy	10 µJ	Completely harmless to the debris
Electrical power consumption	$\approx$ 12 W	Fully suited to the constraints of a 6U CubeSat

— An ultrasensitive SPAD (Single Photon Avalanche Diode) detector. These detectors can detect a single returning photon. The table below summarises the parameters we consider optimal for selecting a suitable detector for our micro-LiDAR [7].

Table 5 – Micro-LiDAR SPAD detector parameters [8].

Parameter	**Specified value**	**Purpose in space**
Detector type	SPAD InGaAs/InP	Sensitive at 1550 nm, single-photon detection
Detection efficiency	$\approx 20\,\%$ at 1550 nm	Detect the weak signal backscattered by the debris
Timing jitter (*jitter*)	≈ 200 ps	Range resolution of a few centimetres
Dark count rate	≈ 10 kHz (cooled detector)	Limit noise to maintain a good signal-to-noise ratio
Operating mode	Gated (*gated*), synchronised with the pulse	Reduce noise and power consumption

— MEMS (Micro-Electro-Mecanical System) micromirrors. These oscillating micromirrors are etched into silicon to steer the beam without heavy mechanical components. The table below summarises the parameters we consider optimal for selecting the mirror assembly for our micro-LiDAR [9].

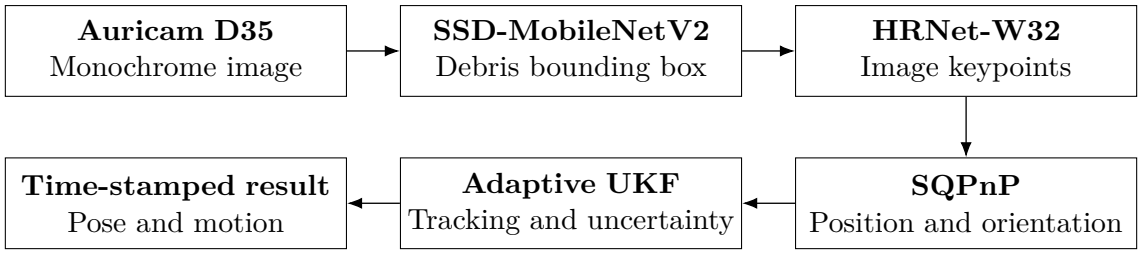
Table 6 – Micro-LiDAR oscillating MEMS micromirror parameters [10].

Parameter	**Specified value**	**Purpose in space**
Scanning mechanism	Resonant MEMS micromirror	Scene scanning without bulky mechanical components
Mirror diameter	≈ 1 mm	Trade-off between optical aperture and resonant frequency
Optical scan angle	$\pm 10°$	Cover the field of view required for target acquisition
Resonant frequency	≈ 1 kHz	Scan rate compatible with real-time tracking
Electrical power consumption	Less than 1 W	Compatible with the power budget of a 6U CubeSat

6 Machine Learning and Intelligent Debris Detection

6.1 Vision Pipeline Organisation

The vision system estimates the **position and orientation of the debris**, then tracks its relative motion. It combines Auricam images, two neural networks and a geometric calculation based on camera calibration and a metrically scaled 3D model of the debris.



Initialisation sequence. During tracking, the UKF uses the observed keypoints and predicts the region of interest in the next image.

The target rate is two optical observations per second, processed as they are acquired. The detector runs at initialisation, during full-image checks and after tracking is lost. Between these calls, the predicted region allows processing to focus on the debris.

6.2 Camera : Monochrome Auricam D35, 25 mm Focal Length

We select the optical head **Sodern Auricam D35**, including its baffle, to capture the outline and geometric features of the debris. Monochrome imaging avoids colour demosaicing ; the global shutter exposes all pixels simultaneously [11].

Characteristic	**Configuration and published specifications**
Sensor	2048×2048 pixels ; pixel pitch of 5,5 μm ; global shutter.
Optics	25 mm focal length ; standard aperture $f/3{,}5$; field of view of 35° diagonally, approximately 25° along each side.
Frame rate	Full frame, 12-bit : 2,5 frames/s. Binning 2×2 : 10 frames/s at 1024×1024.
Interface	SpaceWire, up to 160 Mbit/s.
Power supply	4,6–6 V ; published power consumption of 2 W.
Integration	$80 \times 80 \times 140$ mm including baffle ; mass of 420 g.

The 25 mm focal length combines a useful image scale with a framing margin. A projected dimension of 30 cm at 10 m spans approximately 136 native pixels. The optics and mounting are calibrated, and the focus is then adjusted for the observation range of 2 to 10 m. Pointing complies with the published Sun exclusion angle of 38°.

6.3 Onboard Computing : Q8S, XE2 and Interfaces

The **Xiphos Q8S** receives the images, prepares the network inputs and runs SQPnP, the UKF and data logging. The **CogniSAT-XE2** from Ubotica runs the neural networks on its Myriad X processor [12, 13].

Equipment	Key specifications	Role in Argus
Xiphos Q8S	Zynq UltraScale+ XCZU7EG ; 4 Cortex-A53, 2 Cortex-R5 and FPGA ; 4 GB of error-correcting RAM ; 5 W typical.	Reception, cropping, data preparation, pose estimation and temporal tracking.
CogniSAT-XE2	Myriad X ; PC/104 form factor ; 80 g typical ; 2,5 W typical with a representative network and Ethernet.	Execution of SSD-MobileNetV2 and HRNet-W32 ; return of results to the Q8S.
Carrier board	Power supplies, connectors and interface circuits.	Camera connection and integration of the SpaceWire interface with the Q8S FPGA.

Images arrive via SpaceWire ; the Q8S communicates with the XE2 over Ethernet. Integration verifies the camera data rate, the ARM64 Linux host software and the loading of both networks. Tests measure the latency between exposure and output, as well as the power consumption of the complete system.

The vision system coste around 22 W, including margins and conversion losses, with an allocation of 1 kg and 2U covering the camera, boards and interfaces. These design allocations are separate from the typical power consumption of the equipment.

6.4 Neural Networks : Detection Followed by Keypoint Localisation

SSD-MobileNetV2 receives an image prepared at 320×320 pixels. It searches for the debris and returns a bounding box with a score. This architecture is documented for onboard spacecraft pose estimation [14]. The bounding box selects the region to be analysed.

HRNet-W32 receives this region, extracted from the original image and resized to 256×256 pixels. It locates the keypoints associated with the 3D model : corners, junctions or asymmetric features [15]. The Q8S maps their coordinates back to the native image ; SQPnP initialises the pose, and the UKF then performs tracking.

The networks are trained on the ground using synthetic and real images of the debris under varied lighting conditions. The detector also receives images without debris. Onboard, the weights remain fixed and the models run in FP16. Validation on the XE2 checks keypoint accuracy after compilation, false detections and the processing rate of the complete pipeline.

7 Project Management Tools

Storage (Drive)	Asana(Notion) / Google Drive
Communication	WhatsApp / Google Meet
Task tracking	Asana(JIRA)
Scheduling	Asana(JIRA) / Instagantt(JIRA)
Report preparation	LaTeX

8 Project Organisation

8.1 WBS

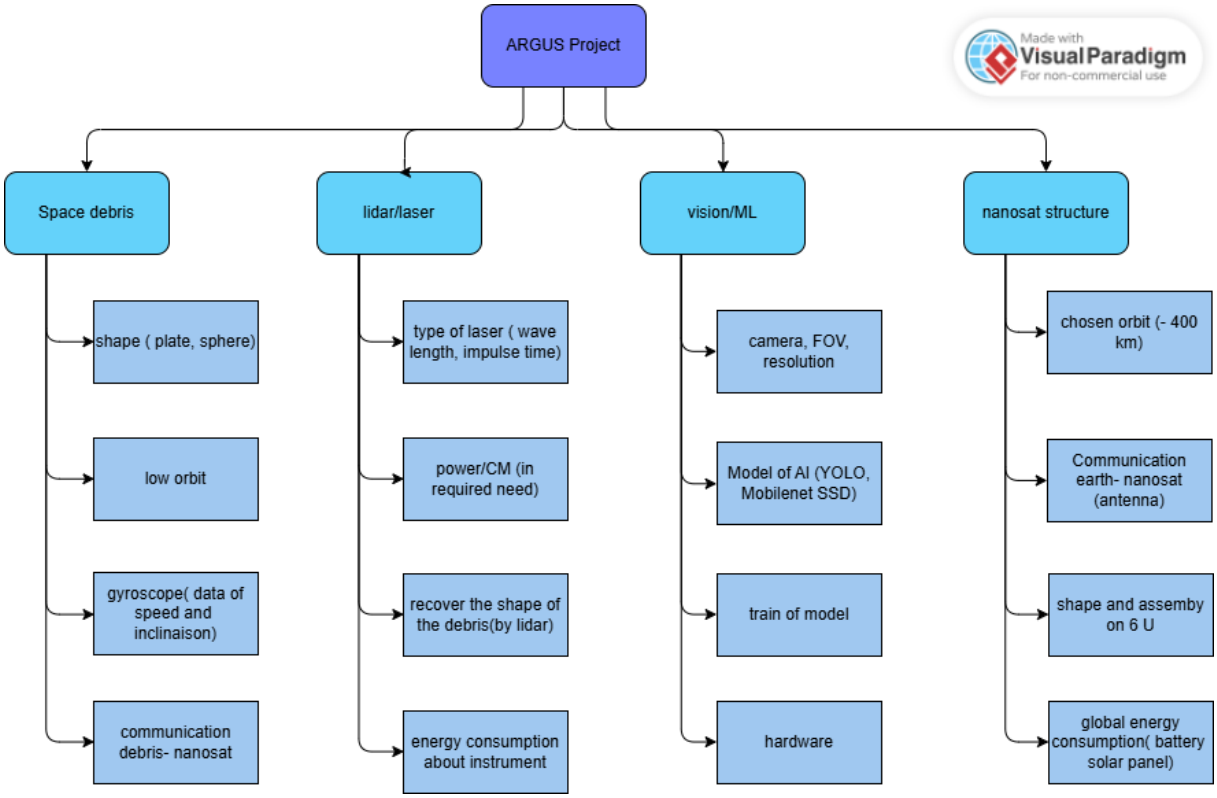


FIGURE 3 – Project WBS

8.2 Roles and Allocated Resources

Name	Role	Time allocation
Adnane AIT ZIDANE	Coordinator and Structure Team Lead	5h/week
Taha EL OUALI	Vision and Machine Learning Team Lead	7h/week
Ahmadou SALL	LiDAR and Laser Team Lead	5h/week
Thomas WEBER	Coordinator and Systems Engineer	7h/week
Brian YOUMBI DIESSE	Systems Engineer	5h/week

8.3 Backward Scheduling

Status : in progress. An issue with our Asana account means that the entire schedule must be recreated on another platform : **JIRA**

8.4 List of Experts

Identified experts	Area of expertise
Ahmadou Sall, Phillipe KECKHUT	LiDAR and laser systems
TBC	orbit
Charlotte Corbelle	Project management

8.5 Risk Analysis

ID	Risk	P	S	Crit.	Mitigation
R1	Poor communication between subteams	3	5	15	Fixed weekly meeting, written minutes, designated contact for each interface
R2	Mass, power or volume budgets exceeded	3	4	12	Maintain an up-to-date budget table ; review at each milestone
R3	Schedule delays and uneven workload distribution	4	3	12	Backward scheduling in Asana, milestones and a 20 % contingency
R4	Temporary unavailability of a team member	4	2	8	Assign a backup for each task ; share documentation
R5	Dependence on a single expert	2	4	8	Summary notes and external experts
R6	Lost or disorganised documents	3	2	6	Single repository, naming conventions and backups
R7	Asana issue	5	2	10	Find an alternative (Notion)

	Severity				
Prob.	**1**	**2**	**3**	**4**	**5**
1					
2				**R5**	**R7**
3		**R6**		**R2**	**R1**
4		**R4**	**R3**		
5					

9 Team MIRO Board

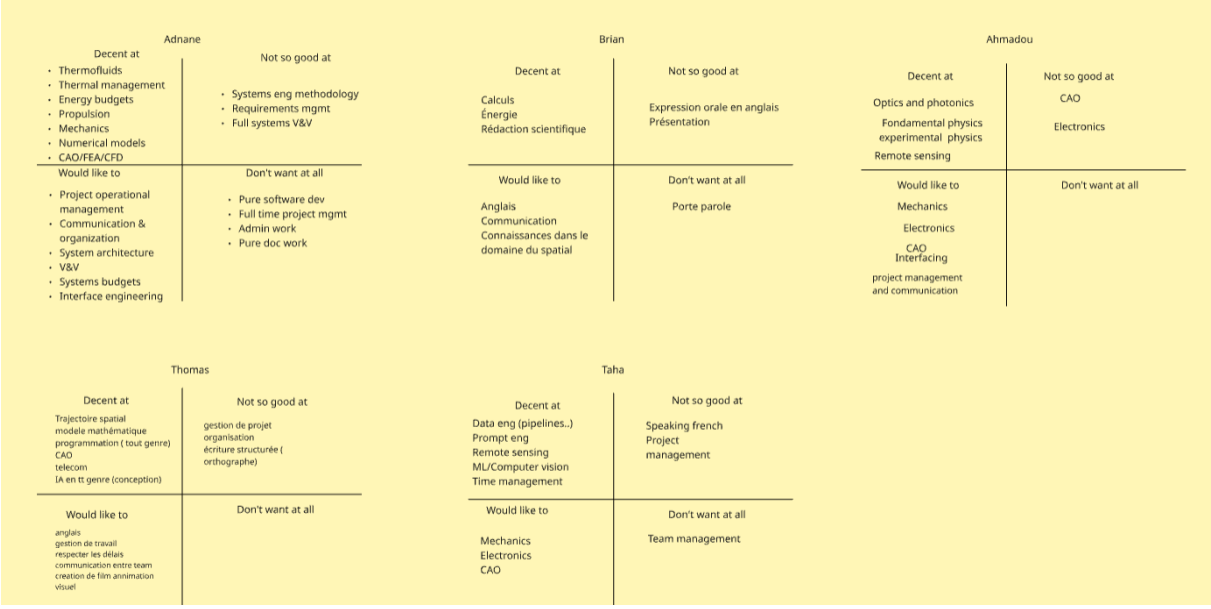


FIGURE 4 – Team MIRO Board

COMPTE-RENDU DE RÉUNION

nom / équipe : **trouvons un nom**

Informations générales

Date	JJ/MM/AAAA
Jour	Lundi / Mardi / Mercredi / Jeudi / Vendredi
Heure de début	HH :MM
Heure de fin	HH :MM
Durée	X heures
Lieu / Visioconférence	Salle / Lien Teams / Zoom / etc.
Responsable du compte rendu	Nom Prénom

Participants

Cochez les personnes présentes :

☐ Adnane
☐ Ahmadou
☐ Brian
☐ Taha
☐ Thomas

Intérêt de la réunion

Décrire en quelques lignes pourquoi cette réunion est organisée, ce que l'on cherche à obtenir et les objectifs principaux.

Objectif principal :

Points discutés

Point 1 — Titre du sujet

Sujet :
Décrire le sujet abordé pendant la réunion.
Discussion :
Résumer les différents éléments discutés.
Décision / conclusion :
Indiquer ce qui a été décidé ou retenu.

Actions à réaliser avant la prochaine réunion

#	**Action à réaliser**	**Responsable**	**Échéance**
1	Action à réaliser	Nom	JJ/MM
2	Action à réaliser	Nom	JJ/MM
3	Action à réaliser	Nom	JJ/MM
4	Action à réaliser	Nom	JJ/MM

Préparation de la prochaine réunion

Date prévue : JJ/MM/AAAA
Heure prévue : HH :MM

Points à traiter lors de la prochaine réunion :

- ☐ Faire le point sur les actions réalisées.
- ☐ Vérifier l'avancement du projet.
- ☐ Traiter le point suivant : .....
- ☐ Prendre une décision concernant : .....
- ☐ Autre : .....

Notes complémentaires

écrire ici les choses en plus

Synthèse de la réunion

Objectif de la réunion	
Décisions principales	
Actions à réaliser	
Responsable(s)	
Prochaine réunion	

Fin de la réunion

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